🚀 RAM Sentinel // Roadmap & Future Architecture

We are building the ultimate privacy-first system optimizer. Many of our Phase 1 & 2 targets were accomplished. Here is the active living blueprint of where RAM Sentinel goes next.

✅ RECENTLY ACCOMPLISHED MILESTONES

- **Hyper-V Game Mode (WAR ROOM):** Completed! Dynamically pauses 30+ background Windows services and de-prioritizes background CPU threads to dedicate 100% of CPU/RAM to the active game.
- **Neural Auto-Sandbox:** Completed! Zero-Trust real-time monitoring that traps and isolates un-whitelisted processes upon excessive memory surges.
- **Deep Memory Compactor:** Completed! Win32 Hooking `psapi` integration natively implemented.
- **System Theme / Widget:** Completed! Built a floating desktop "Dynamic Island" widget and Cyberpunk dashboard UI.
- **VRAM Scanner:** Completed! GPU bottleneck tracking established via WMI.

🧠 Phase 1: Intelligent Automation (The "Neural" Upgrade)

1. Predictive Pre-Caching

- **Concept**: The system learns your daily routine.
- **Use Case**: If you always open Visual Studio Code at 9:00 AM, RAM Sentinel clears 2GB of RAM at 8:59 AM to ensure instant launch times.

- **Tech**: Local-only SQLite database training a lightweight TensorFlow Lite model.

2. Context-Aware Tab Suspension

- **Concept**: Don't just close tabs based on time; understand *context*.
- **Use Case**: Never close a tab that is playing music, compiling code, or has an unsaved form filled out.
- **Tech**: DOM deep-inspection via Playwright.

🌐 Phase 2: Network & Core Security

3. Network Sentinel (Micro-Firewall)

- **Concept**: "Little Snitch" for Windows.
- **Feature**: Real-time graph visualization of which apps are actively "phoning home".
- **Action**: One-click kill switch to temporarily block outbound ports for specific rogue processes.

4. AI Process Fingerprinting (Heuristic Analysis)

- **Concept**: Move beyond static byte thresholds into behavioral modeling.
- **Feature**: Detect zero-day malware or hidden cryptominers natively by analyzing their OS system-call patterns within `psutil` timing arrays.

🔐 Phase 3: Advanced Vault Protocol

5. Steganographic Ghost Vaults

- **Concept**: Hide your Ghost Drive *inside* a harmless looking file on your hard drive before it spins into RAM.
- **Use Case**: To an outsider, it looks like a 4GB `vacation_movie.mp4` (and it plays!). But mount it with RAM Sentinel, and it decodes into a fully functional secure drive.
- **Tech**: AES-256 encryption with offset byte injection.

6. Dead Man's Switch (Anti-Forensics)

- **Concept**: Ultimate hardware data protection.
- **Feature**: If you don't enter a password every X days, or if a specific authorized USB thumb drive isn't plugged in upon booting, RAM Sentinel securely overwrites the memory space and shreds its own configuration files.

☁️ Phase 4: Decentralized Control

7. P2P Encrypted Sync

- **Concept**: Sync your "Read Later" Chrome tabs between your Laptop and Desktop *without* ever using a cloud server.
- **Tech**: WebRTC / Local LAN Sync.
- **Privacy**: Data transfers directly device-to-device. No middleman.

8. Mobile Companion App + Panic Remote

- **Concept**: View your PC's real-time stats securely on your phone over local Wi-Fi.
- **Feature**: Act as a hardware "Panic Button" trigger. One tap on your phone instantly triggers the Emergency Wipe sequence on your desktop.

Verified by Antigravity Design Team (v2.5 Blueprint)